

SUNKARA SIVA SAMBHAVI SANTHOSH

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PROFESSIONAL SUMMARY

Computer Science (AI/ML) student with hands-on experience developing **Machine Learning, Deep Learning, Computer Vision, NLP, Generative AI, RAG, and Reinforcement Learning** solutions. Built and evaluated AI systems including RAG-based LLM chatbot, YOLO/SSD object detection, CNN-based MRI classification, real-time gesture recognition, and NLP spam detection. Proficient in **Python, NumPy, Pandas, scikit-learn, OpenCV, LangChain, Mediapipe, YOLO, and SSD**, with experience in data preprocessing, feature engineering, model training, evaluation, optimization, API integration, real-time inference, and workflow automation. Seeking an **AI/ML Engineer** role focused on developing reliable and scalable machine learning solutions.

TECHNICAL SKILLS

Programming: Python **Machine Learning:** Machine Learning, Supervised Learning, Classification, Feature Engineering, Model Evaluation, Pattern Recognition

Deep Learning: Neural Networks, CNNs, Deep Learning **AI/GenAI:** Generative AI, LLMs, RAG, NLP, Prompt Engineering, Reinforcement Learning

Computer Vision: OpenCV, YOLO, SSD, Mediapipe, Image Processing, Real-Time Object Detection, Real-Time Inference

Data/ML Libraries: NumPy, Pandas, scikit-learn, LangChain **Engineering:** Git, GitHub, API Integration, n8n Workflow Automation

PROFESSIONAL EXPERIENCE

Edunet Foundation — AI/ML Intern Apr 2024 – Jun 2024

- Engineered a real-time hand gesture recognition pipeline using **Python, OpenCV, and Mediapipe** for the SignSpeak accessibility project.
- Optimized spatial feature extraction to reduce inference latency and improve real-time computer vision performance.
- Selected for the **Regional Round** for technical design and project execution.

Google AI/ML Virtual Internship — Virtual Intern Sep 2025 – Nov 2025

- Completed structured training covering **AI/ML concepts, statistical algorithms, and applied modeling techniques**.
- Built and evaluated ML models using structured and unstructured datasets for real-world problem-solving scenarios.

AI/ML PROJECTS

RAG-Based Customer Support Chatbot Dec 2025

- Designed a **Retrieval-Augmented Generation (RAG)** chatbot using LangChain for automated e-commerce customer support.
- Integrated a vector database and embedding model for contextual retrieval and response generation, achieving **90% query-resolution accuracy**.

Stroke Risk Prediction from MRI Scans Sep 2025

- Trained a **CNN** for MRI-based stroke-risk classification, achieving **90% accuracy**.
- Converted model predictions into personalized precaution recommendations, connecting ML output with actionable guidance.

Real-Time Object Detection Engine Mar 2025

- Designed and trained **YOLO and SSD CNNs** for real-time object detection and tracking.
- Processed live video with OpenCV, achieving **80% detection accuracy** under varied lighting and motion conditions.

Sign-Speak: The Silent Communicator Jun 2024

- Built an end-to-end **computer vision pipeline** converting live hand gestures into text and speech in real time.
- Applied coordinate-based frame optimization to process gesture data at **15 FPS**.

Spam Email Detection System Oct 2023

- Developed an **NLP text-classification pipeline** using Naive Bayes and Logistic Regression, achieving **98% accuracy**.

EDUCATION

Kakinada Institute of Engineering and Technology II, Korangi 2022 – 2026

B.Tech – Computer Science & Engineering (AI & ML) **CGPA: 7.44**

GRC Modern Junior College, Ramachandrapuram 2020 – 2022

Intermediate – MPC **60%**

Z.P.P. High School, Draksharama, Ramachandrapuram 2019 – 2020

SSC **80%**

CERTIFICATIONS & ADDITIONAL INFORMATION

Certifications: Google AI/ML Certification; Edunet Foundation AICTE Certification; Prompt Engineering; Python Full Stack Development; Basics of Generative AI; Basics of DevOps.

Languages: English, Telugu, Hindi